

ITA0606 – Machine Learning

Exp-10

Python Program to Implement Expectation-Maximization (EM) Algorithm

Step 1: Import required libraries

```
import numpy as np  
  
import matplotlib.pyplot as plt  
  
from sklearn.mixture import GaussianMixture  
  
from sklearn.datasets import make_blobs
```

Step 2: Generate sample data

```
X, y = make_blobs(  
    n_samples=300,  
    centers=3,  
    cluster_std=1.0,  
    random_state=42  
)
```

Step 3: Display the original data

```
plt.figure(figsize=(7, 5))  
  
plt.scatter(X[:, 0], X[:, 1])  
  
plt.title("Original Data")  
  
plt.xlabel("Feature 1")  
  
plt.ylabel("Feature 2")  
  
plt.show()
```

Step 4: Create Gaussian Mixture Model

```
# EM algorithm is used internally by GaussianMixture
```

```
em_model = GaussianMixture(  
    n_components=3,  
    random_state=42  
)  
  
# Step 5: Train the model using EM algorithm  
em_model.fit(X)  
  
# Step 6: Predict cluster labels  
labels = em_model.predict(X)  
  
# Step 7: Display cluster centers  
print("Cluster Means:")  
print(em_model.means_)  
  
# Step 8: Display covariance matrices  
print("\nCovariance Matrices:")  
print(em_model.covariances_)  
  
# Step 9: Display mixture weights  
print("\nMixture Weights:")  
print(em_model.weights_)  
  
# Step 10: Display log-likelihood  
print("\nLog-Likelihood:")  
print(em_model.lower_bound_)  
  
# Step 11: Display number of iterations
```

```
print("\nNumber of EM Iterations:")
print(em_model.n_iter_)

# Step 12: Display clustered data
plt.figure(figsize=(7, 5))
plt.scatter(
    X[:, 0],
    X[:, 1],
    c=labels,
    cmap='viridis'
)

# Plot cluster centers
plt.scatter(
    em_model.means_[:, 0],
    em_model.means_[:, 1],
    marker='X',
    s=200,
    c='red',
    label='Cluster Centers'
)

plt.title("EM Algorithm - Gaussian Mixture Model")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.legend()
plt.show().
```

OUTPUT

Cluster Means:

```
[[ 4.6  2.1]
 [-2.5  9.0]
 [-6.8 -6.8]]
```

Covariance Matrices:

```
[...]
```

Mixture Weights:

```
[0.333 ... 0.333 ... 0.333 ...]
```

Log-Likelihood:

```
-3.5...
```

Number of EM Iterations:

```
10
```